

Python Modules and File Handling Practice Assignment

Part 1: Modules

1. Create a module:

- Create a Python file named `geometry.py`.
- Inside `geometry.py`, define two functions:
 - `calculate_area_circle(radius)`: Calculates and returns the area of a circle. (Area= πr^2)
 - `calculate_area_rectangle(length, width)`: Calculates and returns the area of a rectangle. (Area=length*width)
 - Import the value of π from the `math` module.

2. Use the module:

- Create a new Python file named `main.py`.
- Import the `geometry` module into `main.py`.
- Prompt the user to enter the radius of a circle.
- Call the `calculate_area_circle()` function from your module, and print the resulting area.
- Prompt the user to enter the length and width of a rectangle.
- Call the `calculate_area_rectangle()` function from your module, and print the resulting area.

Part 2: File Handling

1. Write to a file:

- In `main.py`, prompt the user to enter a list of their favorite fruits, separated by commas (e.g., "apple,banana,orange").
- Open a file named `fruits.txt` in write mode ("w").
- Write the user's input to the `fruits.txt` file.
- Close the file.

2. Read from a file:

- Open the `fruits.txt` file in read mode ("r").
- Read the entire content of the file and store it in a variable.
- Split the string into a list of individual fruits.
- Print each fruit from the list on a new line.

3. Append to a file:

- Prompt the user to enter another fruit.
- Open the `fruits.txt` file in append mode ("a").
- Append a comma and the new fruit to the end of the file.
- Close the file.

4. Read the updated file:

- Open the `fruits.txt` file in read mode ("r").
- Read the entire content and print it to the console.